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NOVELTY ASSESSMENT REPORT

VOICE ACTIVITAED SONG PLAYER

PREPARED FOR	DATE	SEARCH ID
Test Analyst	**20/11/2025**	**cmi72xi3h000**

CONFIDENTIAL & PRIVILEGED

EXECUTIVE SUMMARY

NOVELTY SCORE	PATENTS ANALYZED	RISK LEVEL
0.0%	49	Moderate

This report analyzes 20 patents against the invention features. The analysis reveals significant overlap across many prior art documents concerning voice-controlled music search and playback. Key missing features across the analyzed patents include comprehensive text extraction from voice and detailed voice sample analysis. While several patents touch upon user request interpretation and song search, the integration of all specified invention features, particularly the combination of detailed voice sample analysis with robust text extraction for nuanced user request interpretation and seamless audio playback control, appears to be limited. The overall novelty assessment suggests a 'Not Novel' determination based on the current evidence, with a 'Medium' confidence level due to the scattered nature of feature implementation across the prior art.

INVENTION SCOPE

Search Query

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Voice-controlled music player analyzes voice input for song requests, extracts text, searches for
matches, and plays selected audio.
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Key Technical Features

F1 Voice sample analysis

F2 Text extraction from voice

F3	Song search function
F4	Audio playback control
F5	User request interpretation

Prior Art Stats

TOTAL HITS	RELEVANT CITATIONS	TOP ASSIGNEES
49	**20**	**Multiple entities**

KEY STRENGTHS

- Broad coverage of voice-controlled music search and playback in prior art.
- Many patents demonstrate elements of user request interpretation and song search functionality.

KEY RISKS

No significant risks identified.

STRATEGIC RECOMMENDATIONS

- Focus claim drafting on the synergistic integration of all five features, particularly emphasizing the technical details of text extraction from voice and voice sample analysis.
- Explore novel methods for analyzing voice samples beyond simple recognition for enhanced user request interpretation.

- Consider embodiments that leverage the extracted text and voice sample analysis for unique audio playback control mechanisms.

FEATURE COMPARISON MATRIX

PATENT REFERENCE	F1	F2	F3	F4	F5
KR20010099450	🟢	🔴	🟢	🔴	🟠
DE102023130744	🟠	🔴	🟢	🟢	🟢
JPH1091176	🟢	🟠	🟢	🔴	🟢
JP2002189483	🟢	🟢	🟢	🔴	🟠
TW201042635	🟢	🟠	🟢	🔴	🟢
WO2018023522	🟢	🔴	🟢	🟢	🟠
KR20040074538	🟢	🟠	🟢	🔴	🟢
KR20080000203	🟢	🟠	🟢	🔴	🟢
JP2021033083	🔴	🟢	🟠	🔴	🟠
CN107318036	🔴	🔴	🟢	🔴	🟠
TW201337722	🟢	🔴	🔴	🟢	🟠
US2007208561	🔴	🟢	🔴	🔴	🟠
KR20100086457	🟢	🔴	🟢	🔴	🟠
US2023266873	🔴	🟢	🔴	🔴	🔴
JPWO2014006746	🔴	🟠	🔴	🟢	🔴
JP2014164112	🟢	🔴	🟢	🔴	🟠

KR200374268	Absent	Present	Present	Partial	Partial
KR20080014188	Absent	Present	Present	Partial	Present
CN115203466	Absent	Present	Present	Partial	Present
CN106251866	Absent	Present	Present	Present	Present

Present Partial Absent

FEATURE NOVELTY ANALYSIS

DETAILED PRIOR ART EVIDENCE

Replayer for music files

KR20010099450

Rel: 71%

Date Unkno

ABSTRACT SNIPPET

The present invention provides a playback apparatus that installs a module of voice recognition function and converts a song title input by voice to m...

View Source Document !'

AI ANALYSIS

Patent describes voice recognition for song selection and playback control. Lacks explicit mention of text extraction or user request interpretation beyond song titles.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

INFORMATION PROCESSING DEVICE AND INFORMATION PROCESSING METHOD

DE102023130744

Rel: 70%

Date Unkno

ABSTRACT SNIPPET

An information processing device comprises a control device (101). The control device (101) is configured to detect an utterance of a user, search for...

View Source Document !'

AI ANALYSIS

This patent focuses on detecting user utterances for music playback. It clearly covers user request interpretation and audio playback, with voice sample analysis implied.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Music search device and music playback device

JPH1091176

Rel: 69%

Date Unkno

ABSTRACT SNIPPET

(57) [Summary] [PROBLEMS] To realize a music search device and a music reproduction device capable of selecting a music by inputting a music name or a...

View Source Document!'

AI ANALYSIS

This patent describes a music search device using voice input. It covers voice recognition, song search, and user request interpretation, with partial evidence for text extraction.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Voice input type music search system

JP2002189483

Rel: 69%

Date Unkno

ABSTRACT SNIPPET

(57) [Summary] [PROBLEMS] To provide a voice-input type music search system capable of automatically extracting each search word from a continuous spe...

View Source Document !'

AI ANALYSIS

This patent details a voice input music search system that extracts search words from speech. It covers voice recognition, text extraction, and song search functionality.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

System and method using interactive voice interface to search song according to the human humming or singing sound

TW201042635

Rel: 68%

Date Unkno

ABSTRACT SNIPPET

This invention relates to a system and method using interactive voice interface to search songs in a specific range according to the human humming or ...

View Source Document !'

AI ANALYSIS

This patent describes searching songs using humming or singing. It covers voice analysis for melody matching, song search, and interactive voice interfaces for user requests.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Voice-control music network playback device

WO2018023522

Rel: 68%

Date Unkno

ABSTRACT SNIPPET

A voice-control music network playback device, comprising: an audio signal collection unit (1), a voice recognition and determination unit (2), an inf...

View Source Document !'

AI ANALYSIS

This patent describes a voice-controlled music playback device. It includes voice recognition, parsing commands for track names, and music playback control.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Song selection system and method for servicing song selection

KR20040074538

Rel: 67%

Date Unkno

ABSTRACT SNIPPET

An object of the present invention is to provide a song selection system and a song selection service method that can select a song that best suits th...

View Source Document !'

AI ANALYSIS

This patent focuses on a song selection system using voice. It includes voice input, analysis, and song selection control based on user voice characteristics.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Music file search method using voice recognition

KR20080000203

Rel: 67%

Date Unkno

ABSTRACT SNIPPET

The present invention relates to a music file retrieval method using voice recognition to search for a music file using voice recognition, and to part...

View Source Document !'

AI ANALYSIS

This patent describes a music file search method using voice recognition. It covers voice recognition, feature extraction from voice, and similarity comparison for search.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Input device for karaoke

JP2021033083

Rel: 67%

Date Unkno

ABSTRACT SNIPPET

PROBLEM TO BE SOLVED: To provide a karaoke input device capable of easily requesting an external search system for a music search by voice input. SOL...

View Source Document !'

AI ANALYSIS

Patent describes voice input for music search, including text extraction from voice and user request interpretation. Lacks specific audio playback control or detailed voice sample analysis.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Song search method, smart TV and storage medium

CN107318036

Rel: 67%

Date Unkno

ABSTRACT SNIPPET

The invention discloses a song search method, a smart TV and a storage medium, and belongs to the technical field of smart terminals. The method inclu...

View Source Document !'

AI ANALYSIS

This patent focuses on voice input for song search on smart TVs, including voice recognition and search results display. It partially covers user request interpretation but lacks detailed voice sample analysis or audio playback control.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Music playing device and control method thereof

TW201337722

Rel: 67%

Date Unkno

ABSTRACT SNIPPET

The invention relates to a music playing device and a control method. The music playback device includes a music player, a voice collection device, an...

View Source Document !'

AI ANALYSIS

This patent describes voice control of music playback, including voice recognition, analysis, and execution based on matching voice features. It covers audio playback control and voice sample analysis but not text extraction or song search functions.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Method and apparatus for searching multimedia data using speech recognition in mobile device

US2007208561

Rel: 67%

Date Unkno

ABSTRACT SNIPPET

A method of searching music using speech recognition in a mobile device, the method including: recognizing a speech signal uttered by a user as a phon...

View Source Document !'

AI ANALYSIS

This patent focuses on searching multimedia data using speech recognition in mobile devices. It covers text extraction from voice (phoneme sequence) and partial user request interpretation for search. It lacks specific voice sample analysis or audio playback control.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Song Retrieval Device using Speech Recognition in Karaoke Environment and Its Method

KR20100086457

Rel: 67%

Date Unkno

ABSTRACT SNIPPET

The present invention relates to a semi-period or table of contents for retrieving a title of a song by using a voice recognition technology in a nois...

View Source Document !'

AI ANALYSIS

This patent describes a voice recognition song retrieval device, including voice analysis and speech recognition for selecting songs. It covers voice sample analysis and song search functions. It partially addresses user request interpretation and lacks text extraction or audio playback control.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Audio media playback user interface

US2023266873

Rel: 67%

Date Unkno

ABSTRACT SNIPPET

The present disclosure generally relates to a media playback user interface. In some examples, the media playback user interface enables a search for ...

View Source Document !'

AI ANALYSIS

This patent focuses on an audio media playback user interface that enables searching for text corresponding to speech. It explicitly covers text extraction from voice but lacks evidence for voice sample analysis, song search, audio playback control, or user request interpretation.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Music playback program and music playback system

JPWO2014006746

Rel: 66%

Date Unkno

ABSTRACT SNIPPET

Music is output in accordance with input voice. A voice acquisition step for acquiring an input voice in a computer, and a character string for dete...

View Source Document !'

AI ANALYSIS

This patent describes a music playback program that outputs music based on voice input, including detecting character strings in voice and playing music after a delay. It partially covers text extraction from voice and user request interpretation. It lacks specific voice sample analysis, song search, or audio playback control.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Electrical equipment

JP2014164112

Rel: 66%

Date Unkno

ABSTRACT SNIPPET

The present invention provides an electric device capable of selecting a piece of music suitable for the user and the situation on the spot by acquiri...

View Source Document !'

AI ANALYSIS

This patent provides an electrical device for selecting music based on user voice. It involves voice analysis to generate feature data and collating it to search for music content. It covers voice sample analysis, user request interpretation, and song search functions. It lacks explicit text extraction or audio playback control.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Accompaniment service provision system

KR200374268

Rel: 66%

Date Unkno

ABSTRACT SNIPPET

The present invention relates to an accompaniment service providing system for searching for a song including voice input data as part of lyrics and a...

View Source Document !'

AI ANALYSIS

Patent describes searching songs using voice input, extracting lyrics to text, and controlling playback. It partially covers user request interpretation and audio playback control.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Music file retrieval system and its method in mobile terminal

KR20080014188

Rel: 66%

Date Unkno

ABSTRACT SNIPPET

The present invention relates to a music search service system using speech recognition.

The music file retrieval service system in the mo...

View Source Document !'

AI ANALYSIS

This patent focuses on music retrieval via speech recognition and voice/pitch analysis. It partially covers audio playback control and offers clear song search and text extraction from voice.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

Method, computer storage medium and electronic device for determining audio

CN115203466

Rel: 66%

Date Unkno

ABSTRACT SNIPPET

A method for determining audio, a computer storage medium, and an electronic device, wherein the method for determining audio includes the following s...

View Source Document !'

AI ANALYSIS

This patent describes determining audio via voice commands, identifying keywords, and searching. It covers song search, user request interpretation, and text extraction from voice.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...

A voice-controlled music network playback device

CN106251866

Rel: 66%

Date Unkno

ABSTRACT SNIPPET

The invention relates to a voice-controlled music network playback device, comprising: an audio signal collection unit, a speech recognition judgment ...

View Source Document !'

AI ANALYSIS

This patent details a voice-controlled music playback device, including speech recognition, information analysis, and network search. It covers song search, playback control, and user request interpretation.

FEATURE MAP

- Voice sample...
- Text extract...
- Song search ..
- Audio playba...
- User request...